

Example DOI landing page

Mockup 1

The **first tab** would show the metadata schema documentation, pulled from CEDAR OpenView.

Metadata Reporting Standards

Histology

The microscopic study of tissue composition and structure, often referred to as microscopic anatomy. It involves examining tissue samples, typically after they've been sectioned, stained, and placed under a microscope.

Published: September 26, 2025

<https://doi.org/10.35079/HBM788.QPBW.699>

Subjects: AB-PAS, H&E, H-DAB, LFB, PAS, SBB, Trichrome

Version 1



Download (ZIP)

MD5 hash:

5d41402abc4b2a76b9719d911017c592

Descriptive Metadata

File Organization

Schema

Parent sample ID * <i>Unique HuBMAP or SenNet identifier of the sample (i.e., block, section or suspension) used to perform this assay. For example, for a RNAseq assay, the parent would be the suspension, whereas, for one of the imaging assays, the parent would be the tissue section. If an assay comes from multiple parent samples then this should be a comma separated list. Example: HBM386.ZGKG.235, HBM672.MKPK.442 or SNT232.LBHJ.322, SNT329.ALSK.102</i>
Lab ID <i>An internal field labs can use it to add whatever ID(s) they want or need for dataset validation and tracking. This could be a single ID (e.g., "Visium_90LC.A4.S1") or a delimited list of IDs (e.g., "90L; 90LC.A2; Visium_90LC.A4.S1"). This field will not be accessible to anyone outside of the consortium and no effort will be made to check if IDs provided by one data provider are also used by another.</i>
Preparation protocol DOI * <i>DOI for the protocols.io page that describes the assay or sample procurement and preparation. For example for an imaging assay, the protocol might begin with staining of a section and finalize with the creation of an OME-TIFF file. In this case the protocol would include any image processing steps required to create the OME-TIFF file. Example: https://dx.doi.org/10.17504/protocols.io.eq2lno9qvx9/v1.</i>
Dataset type * <i>The specific type of dataset being produced.</i>
10x Genomics Visium - (https://purl.humanatlas.io/vocab/iravs/0000197)
Analyte class * <i>Analytes are the target molecules being measured with the assay.</i>

Deprecated

Tiled image columns # ☐ <i>The number of columns used in the stitching process of a tiled image, often referred to as the grid size in the x-dimension. Example: 5</i>
Tiled image count # ☐ <i>The total number of raw tiled images captured, which are intended to be stitched together. Example: 75</i>

CEDAR OpenView Documentation

CEDAR Validator

HuBMAP Datasets

The HuBMAP Data Portal is an open platform to discover, visualize and download standardized healthy single-cell and spatial tissue data.

Mockup 2

The **second tab** would show the file hierarchy schema table that's currently in our GitHub documents and the directory schema displayed as a simple tree. Both of these are also included in the directory.md file that's in the DOI zip.

Metadata Reporting Standards

Histology

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Descriptive Metadata

File Organization

Details

pattern	required?	description
extras\/*	✓	Folder for general lab-specific files related to the dataset.
extras\microscope_hardware*.json	✓	[QA/QC] A file generated by the micro-meta app that contains a description of the hardware components of the microscope. Email HuBMAP Consortium Help Desk help@hubmapconsortium.org if help is required in generating this document.
extras\microscope_settings*.json		[QA/QC] A file generated by the micro-meta app that contains a description of the settings that were used to acquire the image data. Email HuBMAP Consortium Help Desk help@hubmapconsortium.org if help is required in generating this document.
raw\/*	✓	Raw data files for the experiment.
raw\images\/*	✓	Raw image files. Using this subdirectory allows for harmonization with other more complex assays, like Velum that includes both raw imaging and sequencing data. This directory should include at least one raw file.

Example

```

.
├── metadata.tsv
├── extras/
│   ├── contributors.tsv
│   └── microscope_hardware.json
└── raw/
    ├── images/
    │   ├── Brightfield_HE_10msec_image_0001.tiff
    │   ├── Brightfield_HE_10msec_image_0002.tiff
    │   ├── ...
    │   └── Brightfield_HE_10msec_image_0110.tiff
    └── lab_processed/
        └── images/
            ├── Histology_13_95.ome.tiff
            ├── Histology_13_95.channels.csv
            └── S1.tissue-boundary.geojson
  
```

HuBMAP Datasets

The HuBMAP Data Portal is an open platform to discover, visualize and download standardized healthy single-cell and spatial tissue data.

Mockup 3

This is the content for the help screen (i.e., the “?” button next to “Download”. It contains descriptive text about each of the files inside the zip archive. This is an incomplete mockup, for the full descriptions, see [here](#).

Metadata Reporting Standards

Documents Provided

The HuBMAP Metadata Reporting Standards include schemas for both descriptive and structural metadata. The descriptive metadata provides a standardized description of samples, experiments, and processing context. The structural metadata provide standardized file organization including hierarchical file and directory specifications, file type and naming conventions

The metadata standards vary for different types of experiments, although there is significant overlap across standards. These standards are reported using five files: JSON (metadata), XLSX (metadata), TSV (metadata), MD (files and directories), ZIP (directories)

metadata.json (JSON file)

JSON Schema Description

This JSON schema defines the structure and validation rules for metadata instances generated from a metadata template in the CEDAR framework. The schema is automatically produced by the CEDAR Workbench and can be accessed through the template's OpenView page.

The schema follows the JSON Schema specification and includes a JSON-LD context to provide semantic meaning for metadata fields. It defines the set of properties, required fields, data types, and value constraints used to validate metadata instances.

Each schema contains:

- Template metadata (name, identifier, version)
- Field and element definitions describing the metadata structure
- Validation constraints such as required fields and controlled values
- UI configuration used to render metadata entry forms

The metadata schema supports the standardized description of HuBMAP datasets by defining a consistent structure for dataset metadata. It enables validation of submitted metadata, promotes interoperability across HuBMAP systems, and facilitates reliable data discovery and reuse.

metadata.xlsx (Microsoft Excel file)

Microsoft Excel File Description

This is placeholder text.
See the Google Doc for the full content.